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Fw: CSCNET S2022 showcase - Thank you!



Jim Pham

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Sat 5/14/2022 7:53 AM

Thank you to those who joined us for the event today! We are so grateful that you made the time in your busy schedules. The students appreciate the opportunity to show their work to all you kindly interested visitors.

Many, many thanks to

- Jim Pham in excellently preparing the CS113 students for this showcase.
- Lori Silverman and Liz Whalley for supporting and solving many logistics of the event.
- Reza Barghi for superbly supporting the students through this semester and making sure today's event goes smoothly.
- Facilities and IT Departments for the great setup and IT support.

I hope everyone is enjoying/enjoyed the Tacos Bravo lunch, provided by the CS/CNET Larry Weiner Fund.

If you couldn't make it this time, hopefully we will see you next time. Some fun photos below of this beautiful day.

Thank you also for your patience with all my many emails today.

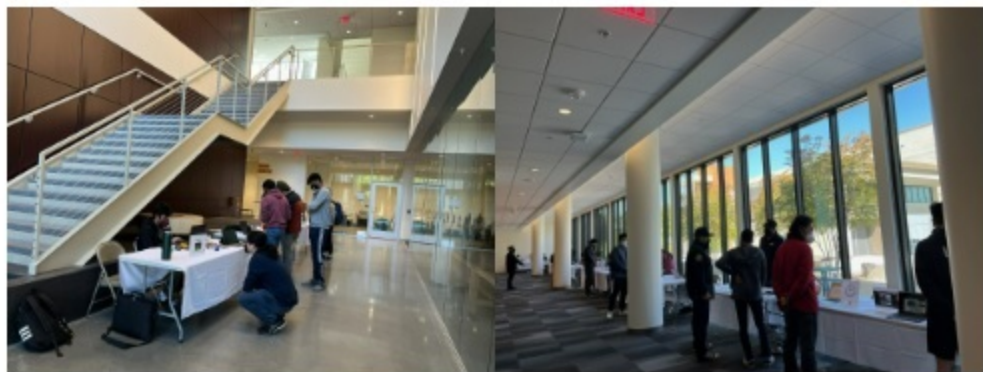
Have a wonderful weekend!

S

Bcc: share@ohlone.edu



It's a beautiful day to meet some students and see some projects!



Arduino Hand Wash Timer

Arlyn Benozza and Kenneth Manuntag



Kalp Patel
Varun Narravula

Drum Hero

(The better Guitar Hero)



Disclaimer: Don't report us for copying. We thought of it first, but they just made it before us

About Us:

Varun - Freshman here at Ohlone

Kalp - Sophomore transferring in Fall

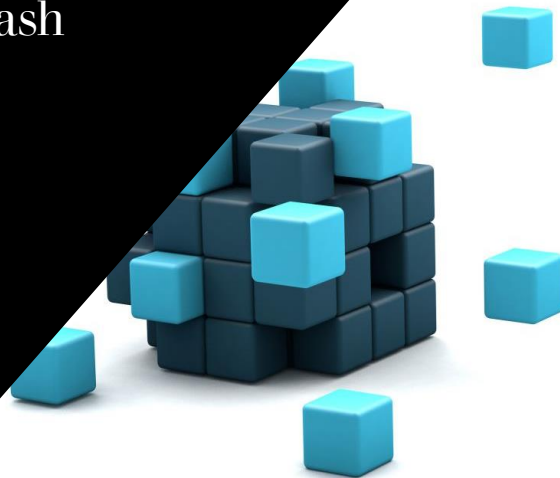
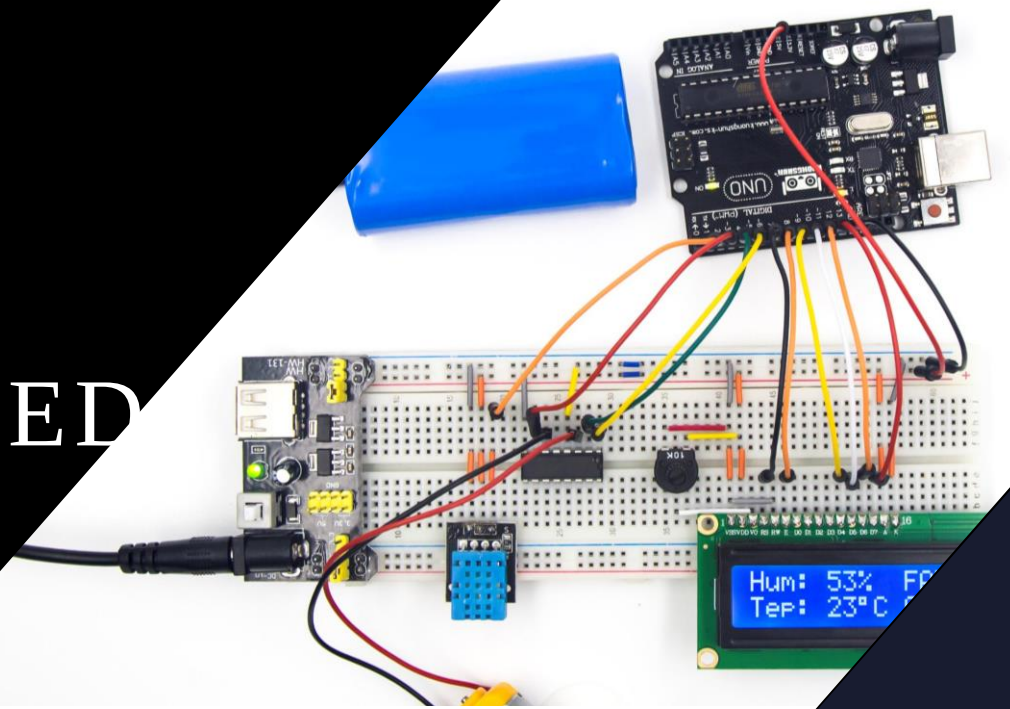
Ask us if you have any questions.

HUMIDITY AND TEMPERATURE CONTROLLED FAN

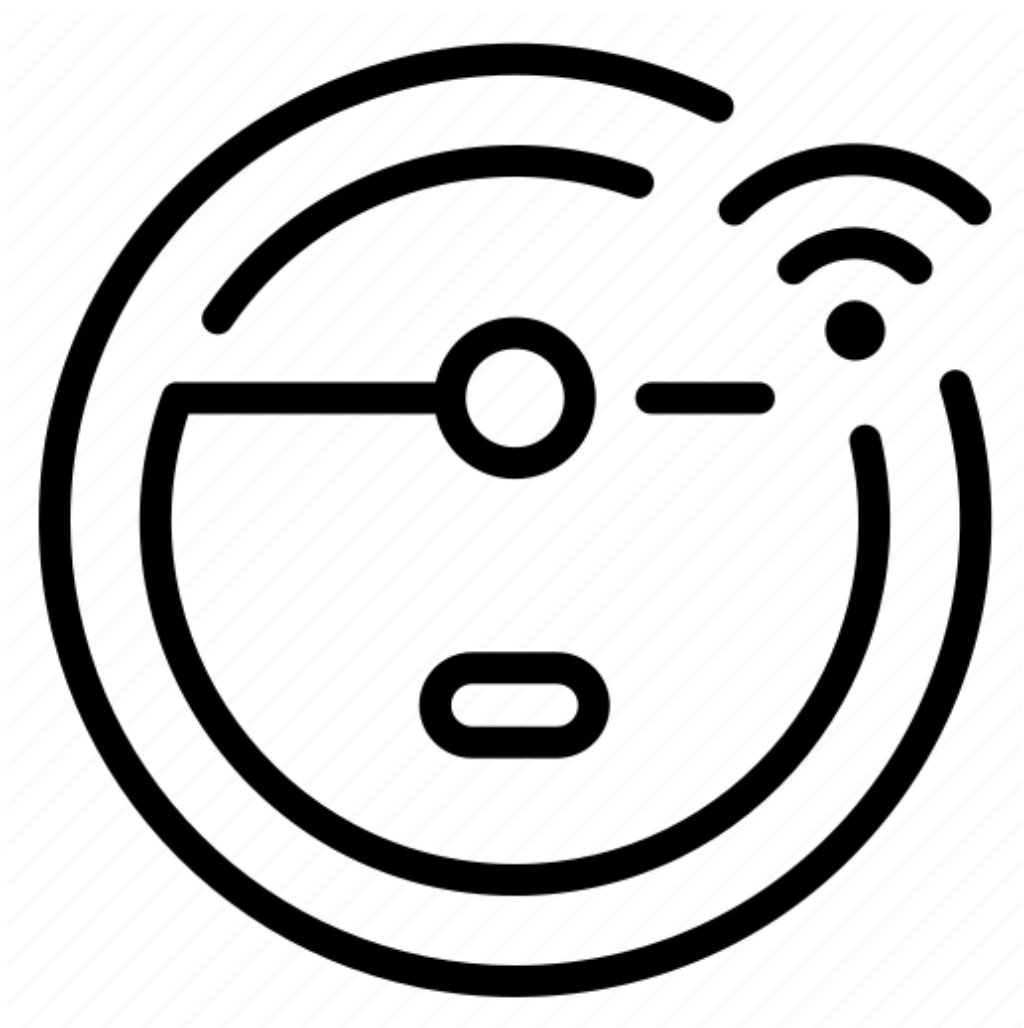
A&P TEAM

Abdulrahman Abuammash

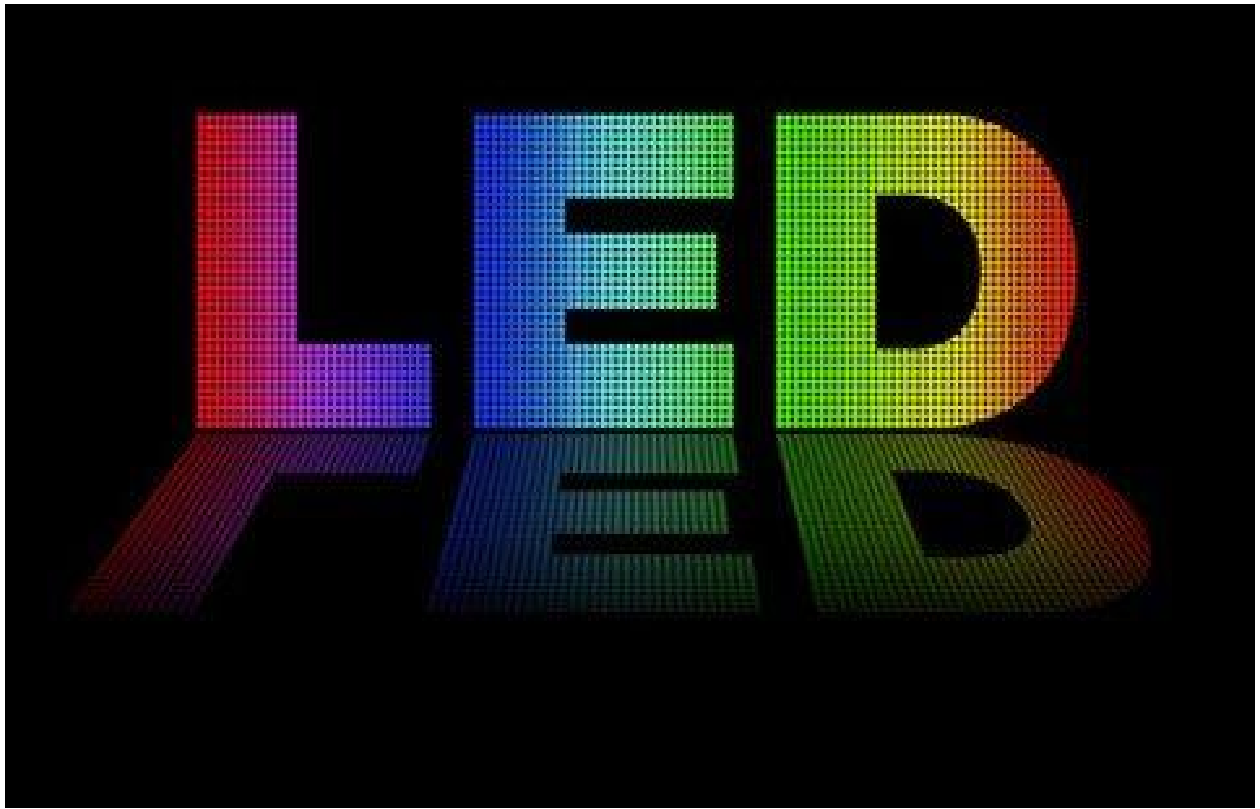
Pramodya Rajapakse



iRobot



Ying Zhou



LED Light Show

Team Shining Lights

05.13.2022

Jotsaroop & Abdarrahan

Between 10:00am - 10:45am

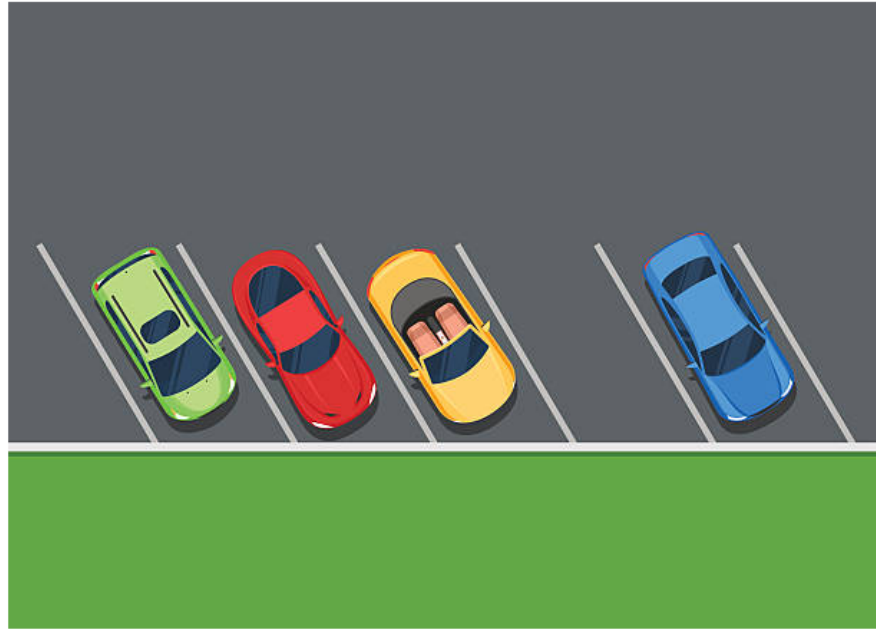
Fremont Campus Building 1, Floor 2

Project Name: LED Strip Light

Mahnoor Kafil
Sadiyah Khan

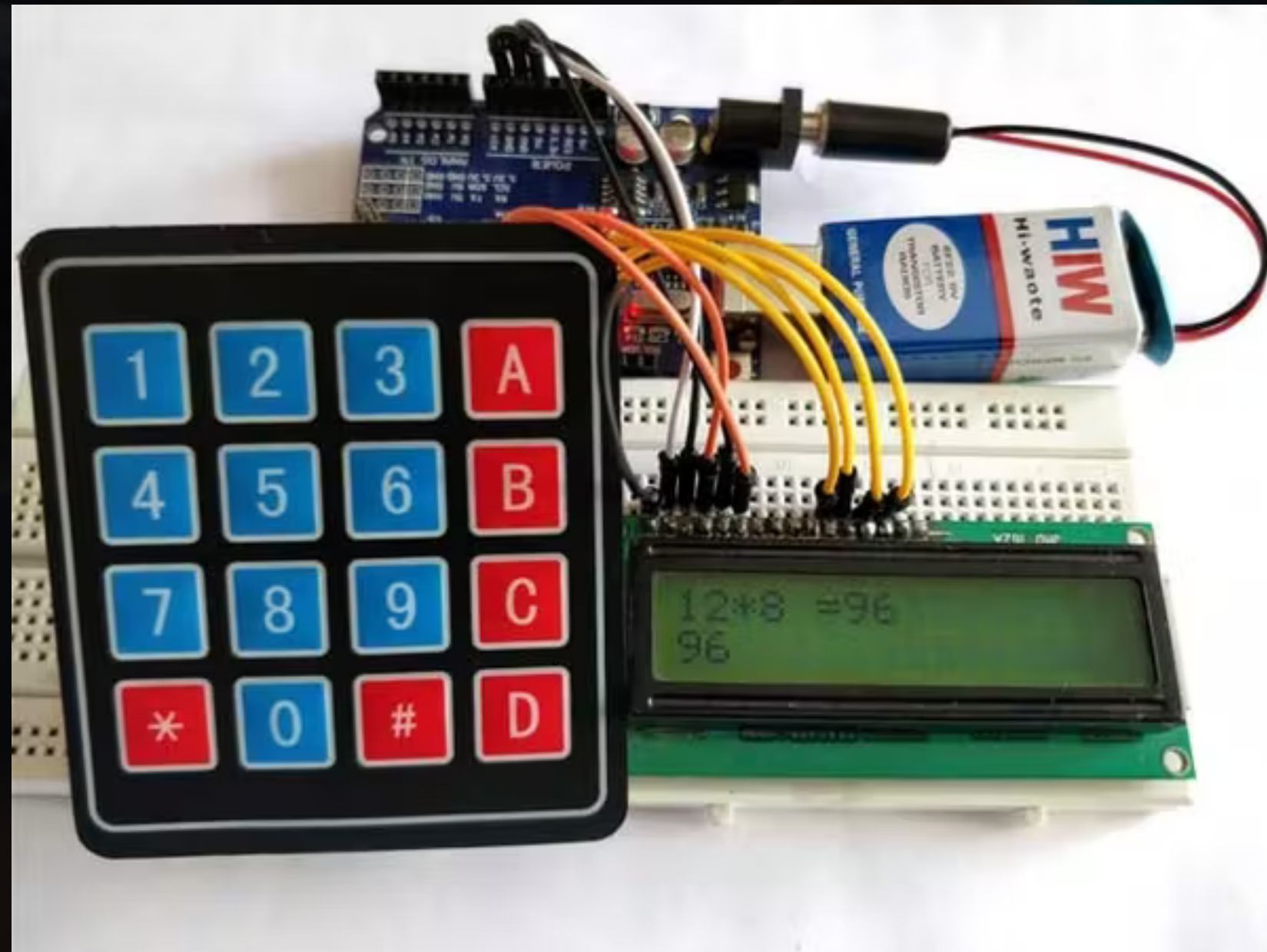


Parking Lot Arduino



Anthony La and Hamza Ahmed

SIMPLE CALCULATOR

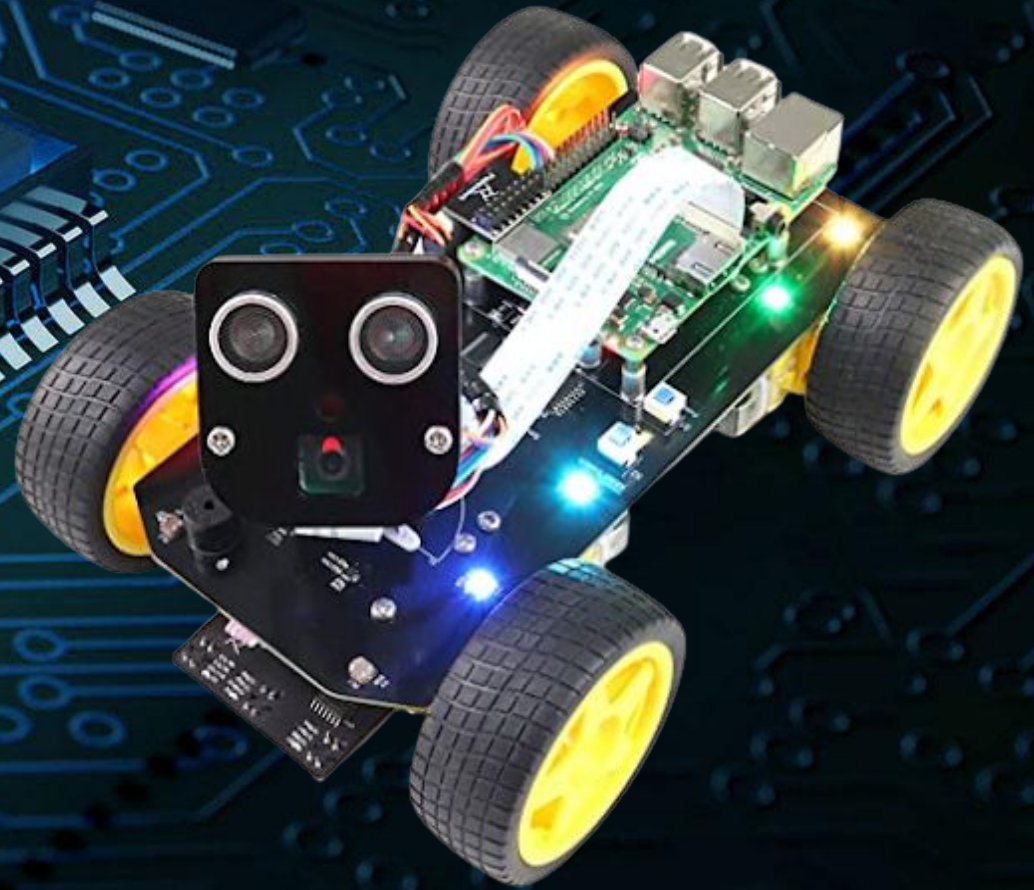


B A O T R U O N G
N G O C H U Y N H

Smart Car Kit with Raspberry Pi Project

Team Speedy

By Su Lei Aye
David Yu



Smart Home Modifier



Team Name: TLM

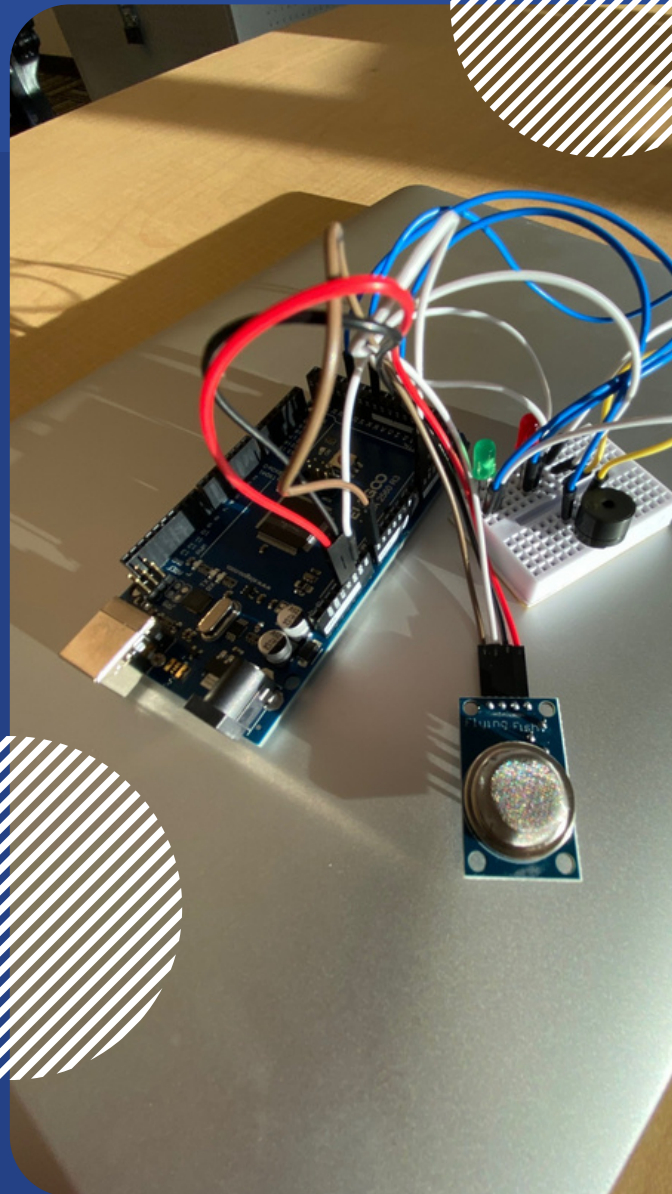
*Christopher
&
Evan*

SMOKE DETECTOR

JASSAN, NHAN

TEAM J&N

USING ARDUINO

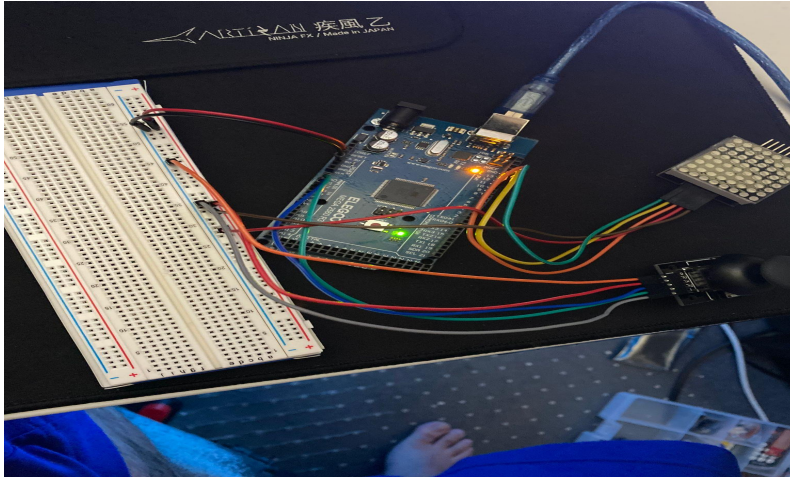




Snake Game

Brandon Villiados
Nicholas Jozovich-McNiel

First ProtoType



- Arduino Uno(Mega 2560 Controller Board),
- Analog joystick (generic),
- Breadboard (generic)
- LED Dot Matrix Display
- Female/Female Jumper Wires
- Male/Female Jumper Wires



Source Code(Struct of the snake)

```
#include <LedControl.h>

//Define The Snake as a Struct
typedef struct Snake Snake;
struct Snake{
    int head[2];    // the (row, column) of the snake head
    int body[40][2]; //An array that contains the (row, column) coordinates
    int len;        //The length of the snake
    int dir[2];     //A direction to move the snake along
};
```

- This shows how the snake starts and constructs throughout the game
- Shows the direction of which way the snake is able to move



Source code cont(struct of the apple)

```
//Define The Apple as a Struct
typedef struct Apple Apple;
struct Apple{
    int rPos; //The row index of the apple
    int cPos; //The column index of the apple
};
```

- This shows how the apple starts on a random row and column on the matrix



Source code cont(LED Matrix)

```
//MAX72XX led Matrix
const int DIN =12;
const int CS =11;
const int CLK = 10;
LedControl lc = LedControl(DIN, CLK, CS,1);

const int varXPin = A3;//X Value from Joystick
const int varYPin = A4;//Y Value from Joystick

byte pic[8] = {0,0,0,0,0,0,0,0};//The 8 rows of the LED Matrix

Snake snake = {{1,5},{0,5}, {1,5}}, 2, {1,0}};//Initialize a snake object
Apple apple = {(int)random(0,8),(int)random(0,8)};//Initialize an apple object

//Variables To Handle The Game Time
float oldTime = 0;
float timer = 0;
float updateRate = 3;
```

- Chose A3 on the Arduino mega 2560 for the x value of the joystick
- Chose A4 on the Arduino mega 2560 for the y value of the joystick
- Snake snake is to get the actual snake on the LED matrix
- Apple apple is to have the apple show up randomly on the LED matrix



Source Code

```
void Update()
{
    reset();//Reset (Clear) the 8x8 LED matrix

    int newHead[2] = {snake.head[0]+snake.dir[0], snake.head[1]+snake.dir[1]};

    //Handle Borders
    if(newHead[0]==8){
        newHead[0]=0;
    }else if(newHead[0]==-1){
        newHead[0] = 7;
    }else if(newHead[1]==8){
        newHead[1]=0;
    }else if(newHead[1]==-1){
        newHead[1]=7;
    }

    //Check If The Snake hits itself
    for(j=0;j<snake.len;j++){
        if(snake.body[j][0] == newHead[0] && snake.body[j][1] == newHead[1]){
            //Pause the game for 1 sec then Reset it
            delay(1000);
            snake = {{1,5},{0,5}, {1,5}}, 2, {1,0}};//Reinitialize the snake object
            apple = {(int)random(0,8),(int)random(0,8)};//Reinitialize an apple object
            return;
        }
    }
}
```

- Check if snake hits itself
 - If true game will pause for “x” amount of time then reset to default settings



Source Code

```
//Check if The snake ate the apple
if(newHead[0] == apple.rPos && newHead[1] ==apple.cPos){
    snake.len = snake.len+1;
    apple.rPos = (int)random(0,8);
    apple.cPos = (int)random(0,8);
}else{
    removeFirst();//Shifting the array to the left
}

snake.body[snake.len-1][0]= newHead[0];
snake.body[snake.len-1][1]= newHead[1];

snake.head[0] = newHead[0];
snake.head[1] = newHead[1];

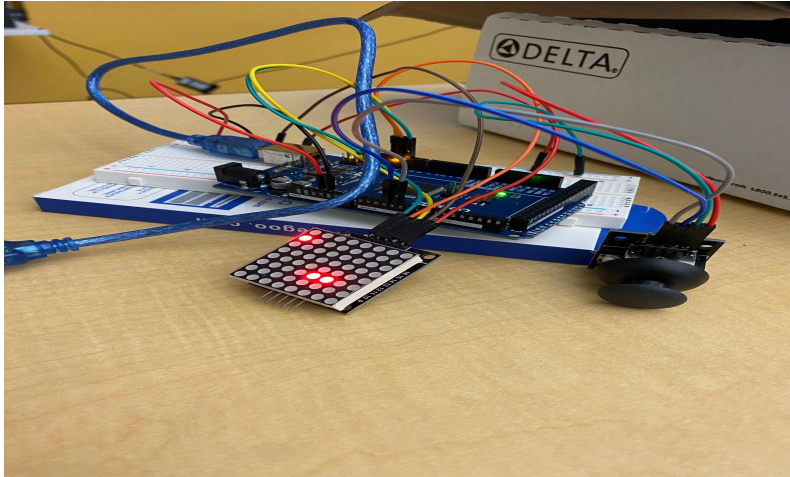
//Update the pic Array to Display(snake and apple)
for(j=0;j<snake.len;j++){
    pic[snake.body[j][0]] |= 128 >> snake.body[j][1];
}
pic[apple.rPos] |= 128 >> apple.cPos;
}

void Render()
{
    for(i=0;i<8;i++){
        lc.SetRow(0,i,pic[i]);
    }
}

void removeFirst()
{
    for(j=1;j<snake.len;j++){
        snake.body[j-1][0] = snake.body[j][0];
        snake.body[j-1][1] = snake.body[j][1];
    }
}
```

- Check if snake eats the apple
 - If true the snake continues to grow
- Shows how the snake and apple are displayed on the LED matrix

Second Prototype



- Got the LED Matrix to work
 - Had issues with the LED control in the libraries previously
- Got the analog joystick to work
- Got the snake game to work correctly



Features and Quantifiers

Function to generate food for snake: domain of food row $-1 < x < 9$. Domain of food column $-1 < y < 9$. Where x and y are random and not generated where snake is generated.

Snake length $z = 3$ where $\text{snake}[\text{pos}(\text{head})] = x = 3$. $\text{Snake}[\text{pos}(\text{tail})] = x = 3-2$. Snake starts at the same position every time in the middle of the matrix where the snake food is randomized but cannot be generated on top of snake.

A stylized green snake is coiled around the text. The snake has a red tongue sticking out, a black eye, and its body forms a large 'G' shape behind the word 'GAME'.

THE SNAKE GAME

Brandon and Niko

Vending State Machine

(By Thomas Sweat)

This program displays a selection of drinks then allows the user to input a code that gives them the corresponding drink.

Drink	Code	Price	Amount left
Coke	a,5	\$1.50	5
Pepsi	b,2	\$1.75	5
Water	d,3	\$1.00	5
7UP	f,7	\$1.25	5

Alternatively, you can input X to terminate the program

Enter a letter: a
Enter a number: 5
You have selected Coke, one bottle will be worth \$1.50
Type in an amount to pay: \$3.46
You are about to purchase 2 bottles of Coke
This will cost \$3.00 with \$0.46 being returned to as change.
Hit any key to continue

This program simulates a vending machine and can let users order drinks. There are a limited amount of drinks and requesting any extra will cause the simulation to give the user back some change. The user is also given back change if they enter an amount of money that is higher than an even amount needed for a specific drink.

The program also features a developer mode that can be accessed by entering the code, "0000#" This mode allows someone to restock existing drinks, add new drinks, or change existing drinks.

Another mode that the program features is a "binary mode" which accepts binary numbers instead of normal characters and can be accessed with the code, "01"

The final feature the program features is an inventory menu that displays what drinks that the user has bought.